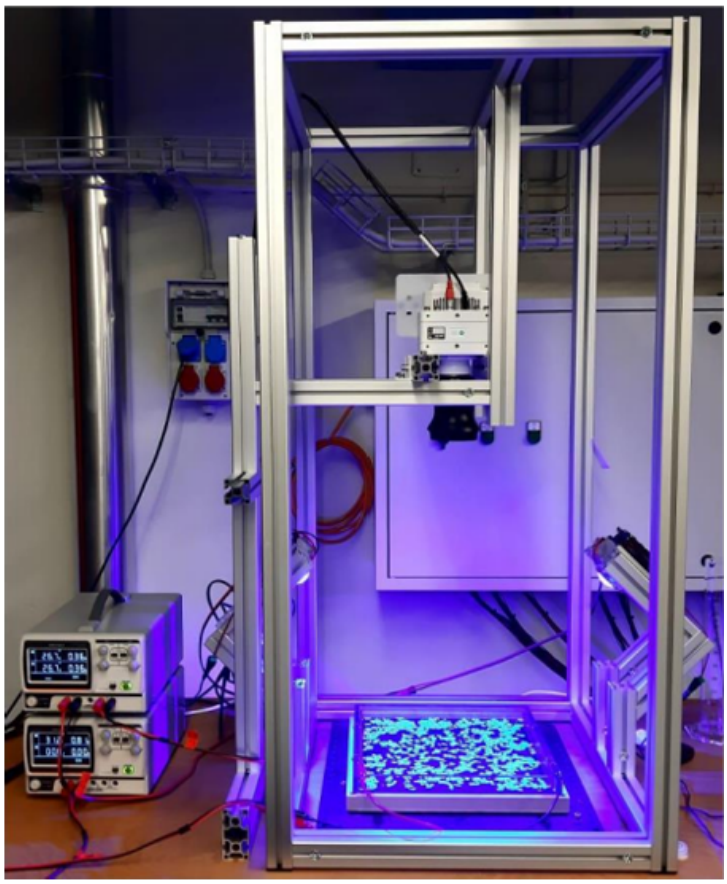
Experiment 2: Particle Tracking Velocimetry in a Thin Fluid Layer

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Experiment: Particle Tracking Velocimetry
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1 Abstract

This experiment investigates a quasi-two-dimensional (Q2D) electromagnetically driven flow using Particle Tracking Velocimetry (PTV). A conductive $CuSO_4$ solution is forced by the Lorentz interaction between an imposed magnetic field and an applied DC current. The image sequences were processed using Particle Tracking Velocimetry to reconstruct individual particle trajectories for four current intensities: 0.01, 0.09, 0.25, and 0.49 A.

From the PTV analysis, the RMS velocity ranges from approximately 6.3×10^{-3} to 3.6×10^{-2} m/s using both boxcar and Gaussian smoothing. The corresponding Reynolds numbers range from 220 to 1270 (boxcar) and from 220 to 1277 (Gaussian), showing an overall increase with the applied current that is broadly consistent with the expected scaling $Re \propto \sqrt{I}$. However, a decrease is observed at 0.49 A, suggesting that experimental limitations may affect the measurements at the highest forcing level. The characteristic flow time scale, $\tau_f = L_f/u_{\mathrm{RMS}}$, decreases from roughly 5.56 s at 0.01 A to about 1.10 s at 0.49 A, reflecting the faster large-scale flow associated with stronger forcing.

Overall, the first three datasets show the expected Lagrangian behavior: velocity PDFs remain close to a Gaussian core, acceleration PDFs exhibit intermittent fluctuations, and the Lagrangian velocity autocorrelation shows decorrelation followed by partial re-correlation at low Reynolds number. The strongest forcing case shows larger deviations, which may be related to shorter particle trajectories at higher velocities and the resulting limitations in statistical convergence.
Keywords: PTV, Q2D flow, Lorentz force, Lagrangian statistics, RMS velocity, Reynolds number, Flow time scale.

2 Introduction

The measurement of fluid velocity has evolved through several techniques, each improving how we observe and quantify flow motion. Early developments relied on Hot Wire Anemometry (HWA), where a small heated wire was placed directly in the flow. Variations in temperature or electrical resistance were used to infer the local velocity. Although effective, HWA is intrusive and only provides point measurements, which limits its use when spatial information is needed. As explained by Bruun [7], these limitations are mainly due to the intrusive nature of hot-wire anemometry.

In the 1970s, Laser Doppler Velocimetry (LDV) introduced a non-intrusive method based on the Doppler shift of light scattered by particles. LDV provides accurate velocity measurements without disturbing the flow, but it remains a point-based technique and requires precise optical alignment. Durst, Melling, and Whitelaw [8] describe in detail how these optical constraints and the focused nature of LDV limit its ability to capture spatial flow structures.

While PIV provides Eulerian velocity fields, its spatial averaging prevents accurate resolution of small-scale particle dynamics such as accelerations. Particle Tracking Velocimetry (PTV) overcomes this by following individual tracer particles in time, providing fully Lagrangian measurements. PTV identifies particle centroids in each frame, potentially with sub-pixel accuracy, and links them across images to reconstruct trajectories. From these, velocities, accelerations, and Lagrangian statistics such as velocity autocorrelation, acceleration PDFs, and mean-squared displacement can be computed. Its performance depends on accurate particle detection, suitable tracer properties, and reliable tracking algorithms. PTV is particularly effective in quasi-two-dimensional (Q2D) flows, where limited out-of-plane motion allows long, continuous trajectories and improved statistical convergence.

In this experiment, a turbulent Q2D flow is generated electromagnetically by driving a current

through a conductive $CuSO_4$ solution above a checkerboard of permanent magnets. By varying the applied current, different Reynolds-number regimes are explored using PTV.

3 Theoretical Background

3.1 PTV Measurement Principle

PTV is a Lagrangian measurement technique, meaning that the fluid motion is described by following individual tracer particles as they move through the flow. Instead of estimating a spatially averaged displacement within interrogation windows, PTV identifies the position of each particle in every frame and reconstructs their trajectories by linking detections across time. The measurement process begins by locating particle centroids in each image. Sub-pixel precision can be achieved by analysing the local intensity distribution, for example through Gaussian fitting. In the Lagrangian description, the flow is described by

$$\mathbf{X}(\mathbf{x}_0, t),$$

which gives the position at time t of the particle that was initially located at $\mathbf{x}_0$. The Lagrangian velocity and acceleration of this particle are then defined as

$$\mathbf{u}(\mathbf{X}(\mathbf{x}_0,t),t) = \frac{\partial \mathbf{X}}{\partial t}(\mathbf{x}_0,t), \qquad \mathbf{a}(\mathbf{X}(\mathbf{x}_0,t),t) = \frac{\partial^2 \mathbf{X}}{\partial t^2}(\mathbf{x}_0,t).$$

PTV provides access to detailed Lagrangian statistics, including velocity autocorrelation, acceleration probability distributions, and mean-squared displacement. In quasi-two-dimensional (Q2D) flows, out-of-plane motion is minimal, enabling long trajectories and improving statistical convergence. Figure 1 illustrates the principle of particle identification and trajectory reconstruction used in PTV.

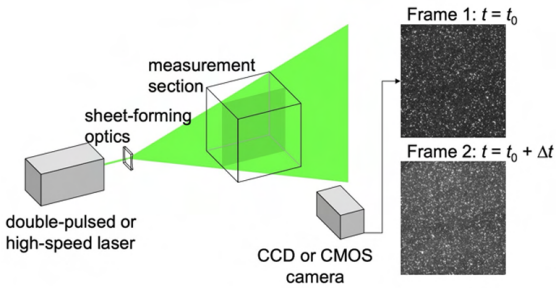


Figure 1: System of pulse laser to capture frames at different instant.

3.2 Lagrangian Velocity Autocorrelation Function

The Lagrangian concept can be compared to a fisherman sitting in a small boat that drifts with the river. Instead of standing on the shore and observing the water flowing past, the fisherman now moves together with the current and experiences how his own velocity evolves over time. In the same way, PTV follows individual particles as they are carried by the flow, providing a trajectory-based description of the motion rather than a field defined at fixed spatial locations.

In the Lagrangian framework, the flow is described by following individual particles along their trajectories. Instead of evaluating how velocity varies at fixed spatial locations , we examine

how a particle remembers its past motion as it is advected by the flow. This memory is quantified using the Lagrangian velocity autocorrelation function (VACF).The normalized Lagrangian VACF for a time lag t is

$$\Gamma_L(t) = \frac{\langle \mathbf{u}_i(t_0) \cdot \mathbf{u}_i(t_0 + t) \rangle_{i,t_0}}{\langle \mathbf{u}_i(t_0) \cdot \mathbf{u}_i(t_0) \rangle_{i,t_0}},$$

where the average is taken over all particles i and over all possible starting times t_0. This correlation measures how long a particle retains information about its initial velocity. The decay of $\Gamma_L(t)$ reflects how long a particle remembers its initial velocity, as described in the PTV manual. A characteristic decay time τ may be defined as the time at which $\Gamma_L(t)$ reaches $1/e$ of its initial value. The manual also emphasizes that all possible time origins t_0 along each trajectory must be used when computing $\Gamma_L(t)$, since relying only on the initial frame of each trajectory leads to poor statistical convergence and noisy correlation curves.

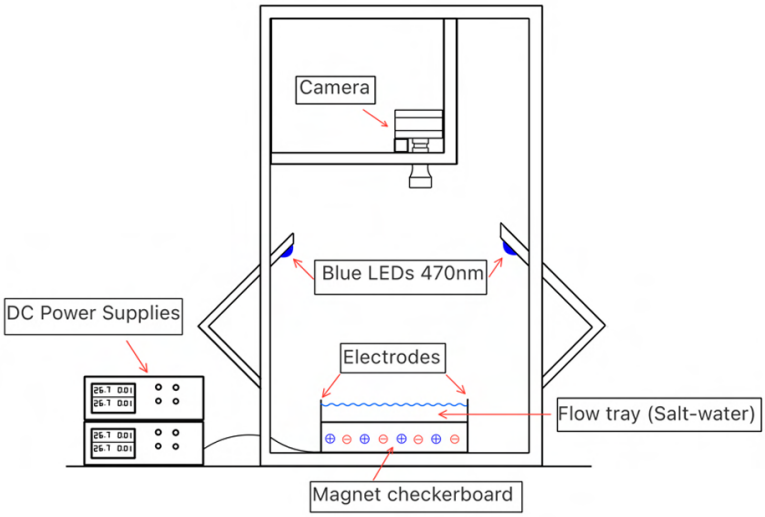


Figure 2: Eulerian perspective of fluid velocity versus Lagrangian depiction of strain [5].

3.3 Physical Variables

3.3.1 Lorentz Force

The electromagnetic forcing applied in this experiment arises from the Lorentz force per unit mass:

$$\mathbf{f}_b = \frac{\mathbf{J} \times \mathbf{B}}{\rho},$$

where $\mathbf{J}$ is the electric current density, $\mathbf{B}$ the magnetic field, and ρ the fluid density. This interaction generates alternating counter-rotating vortices above the magnet array and produces the quasi-two-dimensional flow observed in the experiment.

3.3.2 Reynolds Number and Average Velocity

The characteristic velocity used in this work is the root-mean-square velocity, u_{RMS}. In PTV, the velocity of each particle is obtained from its reconstructed trajectory. The global u_{RMS} is

then computed by averaging the squared velocity magnitudes over all particles and all available time instants. This quantity represents an overall measure of the velocity intensity in the flow.

From theory, the characteristic flow velocity in this type of electromagnetically driven quasi-two-dimensional system is expected to scale as

$$U \approx \left(\frac{NBIL_f^2}{V\rho} \right)^{1/2},$$

which leads directly to the relation

$$Re = \frac{UL_f}{\nu} \approx \frac{L_f^2}{\nu} \left(\frac{NBI}{\rho V} \right)^{1/2}.$$

This shows that the Reynolds number is expected to follow the scaling

$$Re \propto \sqrt{I}.$$

In the Lagrangian framework, the RMS velocity is obtained by averaging the squared particle velocities over all trajectories and all time instants:

$$Re = \frac{u_{\mathrm{RMS}} L_f}{\nu}, \qquad \text{with} \qquad u_{\mathrm{RMS}} = \langle \mathbf{u}_i(t) \cdot \mathbf{u}_i(t) \rangle_{i,t}^{1/2},$$

where the average is taken over all particles i and all time samples along their trajectories, $L_f = 35$ mm is the magnet spacing and $\nu \approx 1$ mm^2/s is the kinematic viscosity of the $CuSO_4$ solution.

3.4 Mean-Squared Displacement

The Mean-Squared Displacement (MSD) describes how far a particle moves away from its initial position over a time interval t. For a particle trajectory $\mathbf{x}_i(t)$, it is defined as

$$\langle r^2(t) \rangle = \left\langle |\mathbf{x}_i(t_0 + t) - \mathbf{x}_i(t_0)|^2 \right\rangle_{i,t_0}.$$

At short times, particles typically move ballistically, leading to a quadratic growth $\langle r^2(t) \rangle \propto t^2$. As velocity correlations decay, the motion becomes diffusive and the MSD grows linearly in time, $\langle r^2(t) \rangle \propto t$. This transition from ballistic to diffusive behavior reflects how particle motion evolves in a turbulent flow.

3.5 Nearest Neighbors and Smoothing

PTV processing consists first in locating tracer particles in each frame by identifying intensity maxima, since tracers appear brighter than the background. Sub-pixel accuracy can be obtained by analysing the local intensity distribution, for example by fitting a 2D Gaussian to the local intensity map. Once particles are detected, trajectories are reconstructed using the nearest-neighbor method, where each particle in frame k is associated with the closest particle in frame $k + 1$. Repeating this procedure over all frames yields individual Lagrangian paths.

To obtain reliable velocities and accelerations from these trajectories, the raw positions must be smoothed. Gaussian smoothing replaces each position by a weighted average of surrounding points using the kernel

$$g(\tau) = \frac{1}{\sqrt{\pi\omega^2}} \exp\left(-\frac{\tau^2}{\omega^2}\right),$$

where points closer in time receive greater weight. Because differentiation commutes with convolution, smoothed velocities and accelerations are obtained by convolving the trajectory with the derivatives of this kernel. The width ω is chosen just large enough to suppress noise while preserving genuine flow dynamics.

A simpler alternative is the boxcar filter, which assigns equal weight to all points within a fixed temporal window. This method effectively removes high-frequency fluctuations but offers less temporal precision than the Gaussian kernel.

For boxcar smoothing, the particle positions are first smoothed using a centred moving average. The velocity and acceleration are then calculated from the smoothed positions using central finite differences. The kernel half-width T determines the number of neighbouring frames included in the smoothing operation, with a total window length of $2T + 1$ frames.

4 Experimental Setup

The measurement system consists of an experimental flow tray with internal dimensions of 300 mm $\times$ 320 mm placed under a high-speed camera mounted on a rigid support. At both ends of the tank, two copper electrodes are installed and connected to a DC power supply, allowing an electric current to pass through the working fluid. Beneath the flow tray, an 8×8 magnetic checkerboard made of neodymium magnets generates an alternating magnetic field. The center-to-center spacing between magnets is $L_f = 35$ mm, which sets the characteristic forcing length scale of the flow.

Two blue LED panels positioned above the tank provide uniform illumination of the flow. The tank is filled with a 10% $CuSO_4$ solution (density $\rho_f = 1080$ kg/m^3) forming a $h = 5$ mm deep fluid layer. Fluorescent green polyethylene microspheres with diameters in the range 75–90 μm and density $\rho_p = 1000$ kg/m^3 are added as tracer particles. Their size and near-neutral buoyancy allow them to follow the flow motion accurately.

Image acquisition is performed using a Baumer high-speed camera. The camera is operated through the Baumer Camera Explorer software, which allows the frame rate, exposure time, region of interest and recording parameters to be adjusted. A checkerboard calibration target is used to convert pixel dimensions into physical units.

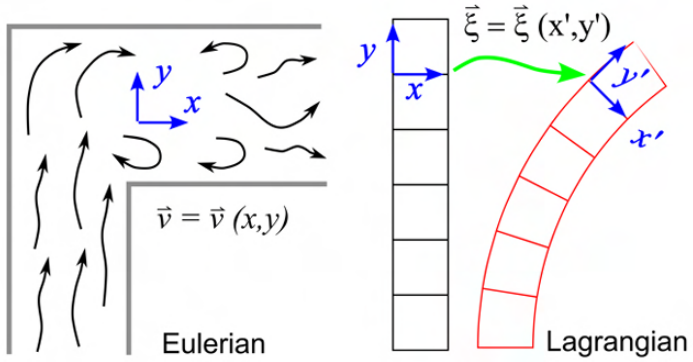


Figure 3: Experimental setup of the Q2D experiment.

5 Methods and Data Processing

5.1 Experimental Procedure

The experiment begins by pouring the premixed 10% $CuSO_4$ solution into the flow tray.

Particle seeding for PTV: In contrast to PIV, where a relatively dense distribution of tracers is required inside each interrogation window, PTV demands a much lower seeding density. Only a small, visually estimated amount of particles is added so that individual tracers remain clearly separated in the images. This ensures that each particle can be uniquely identified and tracked from frame to frame without ambiguity. The fluid is gently mixed before each recording to maintain a uniform distribution of tracers.

Practical steps: The camera is configured with the selected frame rate and exposure time. After waiting briefly for the flow to become fully steady, image sequences are recorded for each applied current. This procedure is repeated for all experimental conditions.

5.2 Calibration and Data Processing

In the previous PIV experiment, a spatial calibration was performed using a checkerboard target placed at the height of the fluid layer. The pixel-to-meter conversion factor was obtained as

$$1 \text{ px} = 8.6514 \times 10^{-5} \text{ m}.$$

The same spatial calibration was used for the PTV measurements, assuming that the camera configuration and measurement plane remained unchanged.

The PTV analysis follows the workflow provided in the 2026 experiment manual. Particle detection is performed in Section 3 of the supplied MATLAB script, where band-pass filtering and particle intensity and size thresholds are adjusted. After checking the detected particle centres, Section 3.b is used to process the image sequence. In Section 4, the nearest-neighbor tracking algorithm links particle detections across consecutive frames to reconstruct individual trajectories. The search radius is adjusted according to the expected particle displacement between frames.

To ensure reliable tracking of each particle, several processing parameters had to be adjusted for every experimental condition.As the applied current increases, larger particle displacements are generally expected between consecutive frames. In addition, thresholds on particle intensity

and acceptable particle radius were imposed during detection in order to filter out noise and non-physical blobs. By retaining only particles within well-defined intensity and size ranges, the risk of incorrect particle detection and trajectory mismatches was reduced.

Table 1: PTV tracking parameters and particle detection thresholds for each applied current.

Current (A)	Search radius (px)	FPS	Intensity range	Particle radius (px)
0.01	5	50	[200-1500]	[1.5-2.5]
0.09	5	100	[100-1200]	[1.5-2.5]
0.25	5	150	[80-1000]	[1.5-2.5]
0.49	5	200	[100-900]	[1.5-2.5]

After the tracking stage, the particle trajectories must be smoothed by selecting an appropriate kernel width. The choice of this parameter is crucial, since an overly narrow kernel fails to remove position noise, whereas an overly wide kernel suppresses genuine flow dynamics. According to theory, one should select the smallest kernel width beyond which the acceleration variance begins to decay exponentially. In this experiment, we follow the procedure proposed by Tropea [1], where the optimal kernel width is identified by examining the evolution of acceleration variance as a function of smoothing strength as we can see on the figure below.

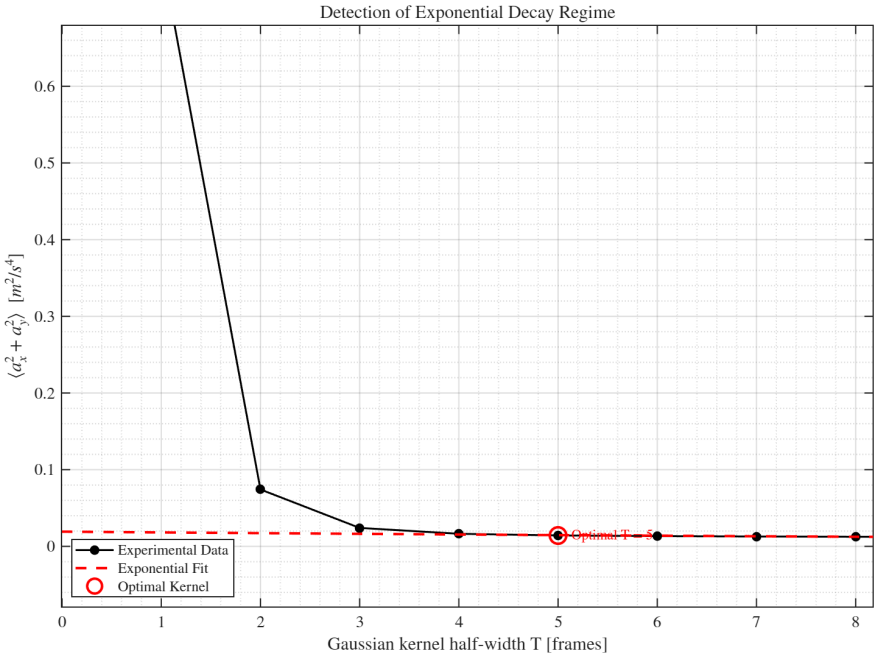


Figure 4: Normalized acceleration as a function of kernel width normalized by τ_η for the 0.25A regime.

6 Results

6.1 Trajectories and Eulerian Conversion

The particle trajectories, along with the reconstructed velocity and vorticity fields, are presented in Section 9. These visualizations allow us to observe how the flow structure evolves as the applied current increases, and how the Lagrangian particle motion reflects the transition from an ordered regime to a more turbulent one.

Trajectory Visualization

At low current, the trajectories are long, smooth and well organized, reflecting coherent particle motion in a laminar or weakly perturbed regime. As the forcing increases, the trajectories generally become shorter and more irregular. This behaviour is consistent with the challenges of particle tracking in turbulent flows discussed by Ouellette et al. [2], where strong velocity fluctuations make the reconstruction of continuous trajectories more difficult. In the highest current case, Figure 18 shows particularly short and fragmented trajectories, producing paths that visually resemble "confetti".

Discussion

The parameters `bplo`, `bphi` and the search radius strongly influence the recovered trajectory lengths, since inappropriate values can reduce the ability to detect and track particles reliably. The importance of accurate particle identification and matching has also been discussed in early PTV studies by Maas et al. [3]. Extensive testing showed that adjusting these parameters within physically reasonable ranges did not substantially improve the trajectory lengths at the highest forcing level. The severe reduction observed at high current may therefore be associated with stronger velocity fluctuations, larger instantaneous particle displacements and possible out-of-plane motion. However, limitations of the detection and tracking algorithms cannot be completely excluded.

Regarding the Eulerian fields, the agreement between PIV and PTV is good for the lowest current. In this regime, the flow appears relatively organized and sufficiently steady for both methods to capture similar velocity and vorticity patterns. At higher forcing, however, differences appear between the PIV and PTV fields. This behaviour can partly be explained by the different processing methods. PIVlab reconstructs an instantaneous Eulerian field from an image pair, while the Eulerian conversion applied to PTV data averages particle velocities onto a spatial grid using multiple trajectories and time instants. This averaging smooths rapid temporal variations and combines information from particles passing through the same location at different times. As the flow becomes increasingly unsteady, the two representations may therefore differ more significantly. Differences in spatial resolution, particle sampling and interpolation may also contribute to the observed discrepancies.

Uncertainties and Performance Estimation

The ability to track particles reliably also depends on the temporal resolution. The performance criterion introduced in the lecture is $\xi = u\Delta t/d_{NN}$, where values below 0.1 are generally recommended to limit mismatches in nearest-neighbor tracking. The inter-particle spacing d_{NN} depends on the seeding density. Particle concentration and image quality are important factors in velocimetry measurements, as discussed by Santiago et al. [4].

Assuming approximately ten particles per interrogation window gives an estimated inter-particle spacing of $d_{NN} \approx 2.5$ mm. Using the Gaussian-smoothed value $u_{\mathrm{RMS}} \approx 31.9\ \mathrm{mm\,s^{-1}}$ at 0.49 A leads to

$$(\Delta t)^{-1} > \frac{u_{\mathrm{RMS}}}{0.1\, d_{NN}} \approx \frac{31.9}{0.25} \approx 128 \text{ fps}.$$

Our acquisition frame rate of 200 fps satisfies this estimated requirement. The corresponding tracking parameter is

$$\xi = \frac{u_{\mathrm{RMS}}}{f\, d_{NN}} \approx \frac{31.9}{200 \times 2.5} \approx 0.064 < 0.1.$$

Two additional sources of uncertainty must be considered. First, the actual particle spacing

may have been smaller than assumed, which would increase the tracking parameter and the required frame rate. Second, the RMS velocity does not represent the largest instantaneous particle velocities, meaning that some particles may have experienced larger values of ξ than estimated.

Although the acquisition frequency satisfies the criterion based on the RMS velocity and the assumed particle spacing, this does not guarantee successful tracking under all conditions. The reduced trajectory lengths observed at high current may therefore result from a combination of stronger velocity fluctuations, possible out-of-plane motion and limitations in particle detection and matching. A higher acquisition frame rate or improved tracking methods could help determine the relative importance of these effects, although this would increase the amount of data to process.

6.2 Reynolds Numbers

In the previous section, the PIV analysis confirmed the theoretical scaling $Re \propto \sqrt{I}$ for electromagnetically driven quasi-two-dimensional flows. The experimentally measured Reynolds numbers followed this relation closely, indicating that the characteristic velocity indeed grows with the square root of the applied current. Having established this behavior using Eulerian measurements, and as discussed in Section 3.3.2, we now turn to the Lagrangian results obtained from PTV. By computing the RMS velocity from particle trajectories and applying the same definition $Re = u_{\mathrm{RMS}} L_f / \nu$, we can assess whether the Lagrangian measurements produce Reynolds numbers consistent with the PIV results and with the expected theoretical trend. The values of u_{RMS} and Re obtained using both boxcar and Gaussian smoothing are summarized below.

Table 2: Comparison of PTV results using Boxcar and Gaussian smoothing.

	Boxcar		Gaussian	
Current (A)	u_{RMS} (m/s)	Re	u_{RMS} (m/s)	Re
0.01	0.006296	220.34	0.006300	220.50
0.09	0.023631	827.07	0.023725	830.37
0.25	0.036291	1270.19	0.036494	1277.30
0.49	0.031618	1106.65	0.031911	1116.88

For completeness, the characteristic flow time scale can be estimated as

$$\tau_f = \frac{L_f}{u_{\mathrm{RMS}}},$$

which represents the time required for a fluid structure to travel across one forcing cell. Using $L_f = 35$ mm and the Gaussian-smoothed RMS velocities for the four forcing levels, we obtain

$$\tau_{f,0.01\mathrm{A}} \approx 5.56 \text{ s}, \quad \tau_{f,0.09\mathrm{A}} \approx 1.48 \text{ s}, \quad \tau_{f,0.25\mathrm{A}} \approx 0.96 \text{ s}, \quad \tau_{f,0.49\mathrm{A}} \approx 1.10 \text{ s}.$$

These values quantify how rapidly the large-scale flow evolves and help assess the temporal resolution needed to resolve particle motion reliably.

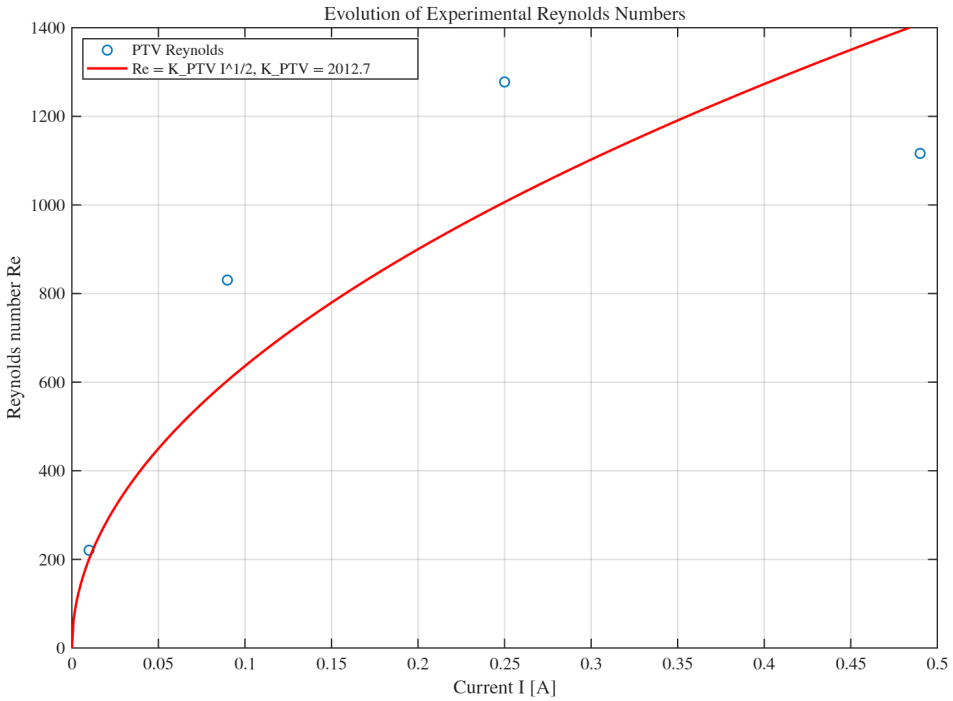


Figure 5: Evolution of the experimental Reynolds numbers compared to PTV theoretical scaling

Discussion: We observe that the Reynolds numbers obtained from PTV lie above the theoretical curve derived from the PIV analysis using $K \approx 2792$ for the first three current values, while the last point falls below it. The deviation is most pronounced at the highest current, where the largest particle displacements and the stronger turbulence may make Lagrangian measurements more sensitive to tracking errors and smoothing. Nevertheless, the four PTV data points show an overall trend consistent with the expected $\sqrt{I}$ scaling, despite the deviation at the highest current. By fitting the relation $Re = K_{\mathrm{PTV}}\sqrt{I}$ to the PTV results, we obtain a new proportionality factor $K_{\mathrm{PTV}} \approx 2014$, indicating that the Lagrangian Reynolds number remains broadly consistent with the expected square-root dependence on the applied current.

The differences between Boxcar and Gaussian smoothing remain very small, indicating that both filters suppress high-frequency noise to a comparable degree. The Gaussian-smoothing results are consistently slightly higher than the Boxcar results, although this does not necessarily indicate a better agreement with the theoretical trend. Gaussian convolution provides a weighted average that gives more importance to points near the evaluation time than the uniform Boxcar filter, as discussed in Tropea's treatment of smoothing kernels [1] and in the PTV manual. Both methods therefore provide similar estimates of the RMS velocity and Reynolds number, although their influence on acceleration estimates may be more significant.

The table below compares the RMS velocities and Reynolds numbers measured using PIV and PTV for the same forcing conditions.

Table 3: Comparison between Eulerian (PIV) and Lagrangian (PTV) measurements of u_{RMS} and Re.

	PIV		PTV (Gaussian)	
Current (A)	u_{RMS} (m/s)	Re	u_{RMS} (m/s)	Re
0.01	0.00702	246	0.0063001	220
0.09	0.0284	993	0.0237249	830.372
0.25	0.0471	1650	0.036494	1277.3
0.49	0.0489	1711	0.031911	1116.9

6.3 Velocity and Acceleration Probability Distribution

For the PTV analysis, the probability density functions of velocity and acceleration are computed directly from the smoothed particle tracks. For each dataset and each current, the Lagrangian velocity components (v_x, v_y) and accelerations (a_x, a_y) of all particles and all time steps are gathered into a single ensemble. Fluctuations are obtained by subtracting the mean value of each component, and they are normalized by their respective standard deviations, yielding dimensionless variables

$$u'_{\mathrm{norm}} = \frac{u'}{\sigma_{u'}}, \qquad a'_{\mathrm{norm}} = \frac{a'}{\sigma_{a'}}.$$

The normalized samples from both components are then combined, and smooth PDFs are estimated using a kernel density estimator over a common symmetric range. The resulting distributions are finally compared to a standard Gaussian $\mathcal{N}(0, 1)$, using a logarithmic scale on the vertical axis to highlight the behavior of the tails for the different flow conditions and smoothing methods.

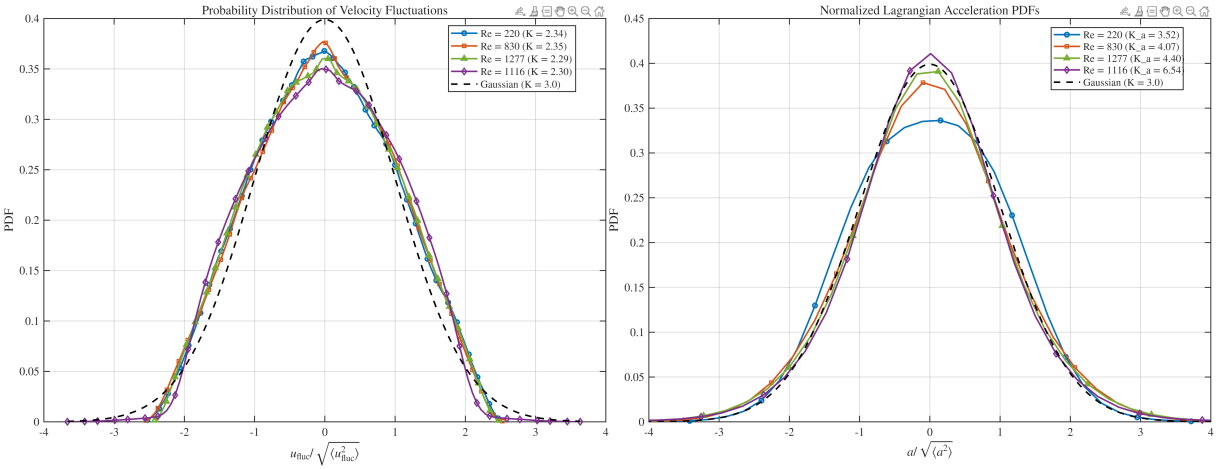


Figure 6: Probability distribution of the velocity magnitude and the normalized velocity fluctuations.

Discussion: In Lagrangian turbulence, the normalized PDF of velocity fluctuations is often approximately Gaussian, while the acceleration PDF is known to remain strongly non-Gaussian with a sharp peak at zero and heavy tails, as demonstrated by La Porta et al. in their landmark experiment [6]. The first three datasets follow these tendencies reasonably well, showing a clear symmetric peak in the velocity PDFs and broader acceleration distributions. However, the fourth dataset shows deviations from these expected behaviors. As discussed previously, the

trajectories at high current are extremely short, which may reduce the number of statistically independent samples and affect the convergence of the PDFs. These tracking limitations may contribute to the distorted shapes, although their influence cannot be separated conclusively from possible changes in the flow dynamics.

6.4 Lagrangian Velocity Autocorrelation and Mean Square Displacement

As discussed in Section 3.2, the Lagrangian velocity autocorrelation function quantifies how long a particle retains memory of its initial velocity while it is carried by the flow. For each trajectory, the inner product $\mathbf{u}(t_0) \cdot \mathbf{u}(t_0 + t)$ is computed for all possible starting times t_0, and the correlation is normalized by the mean squared velocity. Averaging over all particles and all time origins produces the Lagrangian correlation $\Gamma_L(t)$, whose decay characterizes the temporal coherence of the flow.

In parallel, the mean square displacement (MSD) describes how far particles move away from their initial position. The MSD typically exhibits an initial ballistic regime proportional to t^2, followed by a linear diffusive regime as velocity correlations decay. Together with $\Gamma_L(t)$, the MSD provides a complementary view of how turbulence influences particle dispersion.

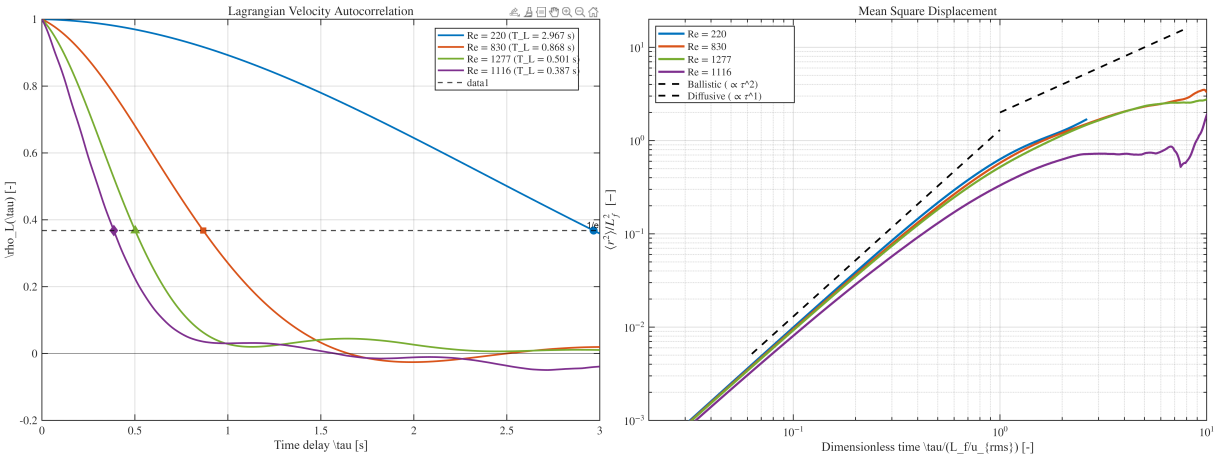


Figure 7: VACF and MSD evolution

Discussion: For the lowest current, the Lagrangian velocity autocorrelation does not decrease monotonically. After an initial drop, the curve increases again, showing that particle velocities regain some correlation at later times. This behaviour reflects the presence of coherent flow structures that persist long enough for particles to experience similar motions after a certain delay. As the forcing increases, this re-correlation becomes less pronounced and the decay generally becomes faster, suggesting that velocity fluctuations lose coherence more rapidly. In the highest forcing case, the correlation decreases rapidly and shows little recovery, which may reflect stronger velocity fluctuations, although the shorter particle trajectories can also affect the reliability of the autocorrelation estimates.

The mean square displacement exhibits the corresponding behaviour. At short times, the MSD grows approximately proportionally to t^2, representing the ballistic regime. As the velocity correlation decays, the MSD may progressively approach a linear growth in time, characteristic of diffusive particle spreading. This evolution is consistent with the loss of velocity correlation observed in the autocorrelation curves, although the limited trajectory lengths make the identification of a fully developed diffusive regime more uncertain.

7 Conclusion

This second part of the experiment applied Particle Tracking Velocimetry to the same quasi-two-dimensional electromagnetically driven flow, with the aim of characterizing its Lagrangian dynamics across four forcing levels. From the particle trajectories, we extracted the RMS velocities, Reynolds numbers, velocity and acceleration probability distributions, Lagrangian velocity autocorrelation functions, and mean square displacements.

While the general trend $Re \propto \sqrt{I}$ is still observed, the quantitative agreement with the PIV-based scaling deteriorates at high forcing. This discrepancy may partly originate from the rapid loss of trackability at higher forcing levels, where particles can only be followed over very short frame lengths. As a consequence, some of the Lagrangian statistics may be affected by limited statistical convergence. Similarly, for the fourth dataset, the Lagrangian velocity autocorrelation decays rapidly without clearly exhibiting the expected dynamical features, as illustrated in as illustrated in the Appendix. These limitations may be linked to tracking uncertainties, stronger velocity fluctuations and the reduced number of statistically independent samples available in the most strongly forced case.

It would therefore be valuable to repeat the experiment with a higher acquisition frequency, for example 300 or 400 fps, in order to investigate whether smaller inter-frame displacements improve trajectory reconstruction. Comparing the current datasets with higher-fidelity PTV measurements would help clarify how much of the deviation comes from physical effects and how much is due to tracking and sampling limitations. Additional experiments with improved seeding control or multi-camera setups could also provide more reliable acceleration statistics and a clearer view of Lagrangian dispersion in Q2D turbulence. Such extensions would allow a more robust comparison between Eulerian and Lagrangian descriptions of the same flow and offer deeper insights into the mechanisms governing the transition from coherent to turbulent dynamics in electromagnetically driven layers.

References

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8 Appendix

First mode: 0.01A

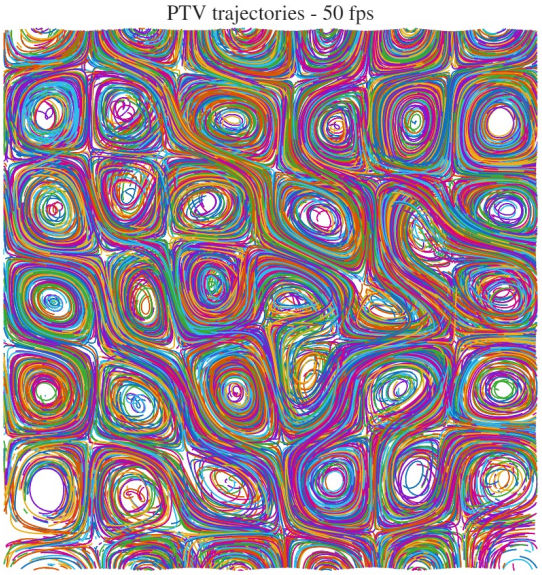


Figure 8: PTV Particle tracing.

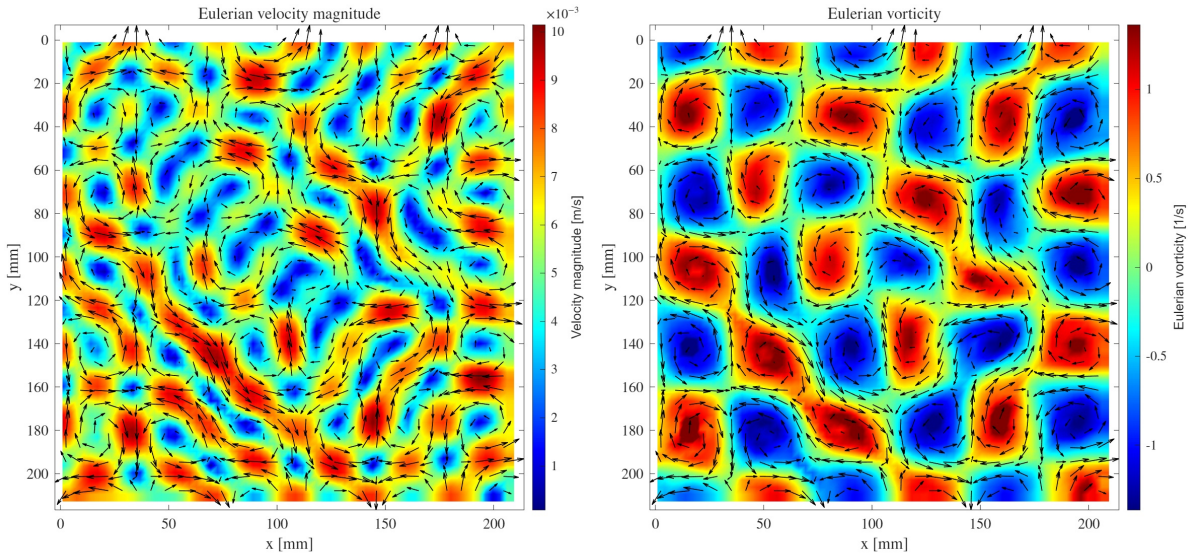


Figure 9: Eulerian conversion map of Vorticity and Magnitude

Second mode: 0.09A

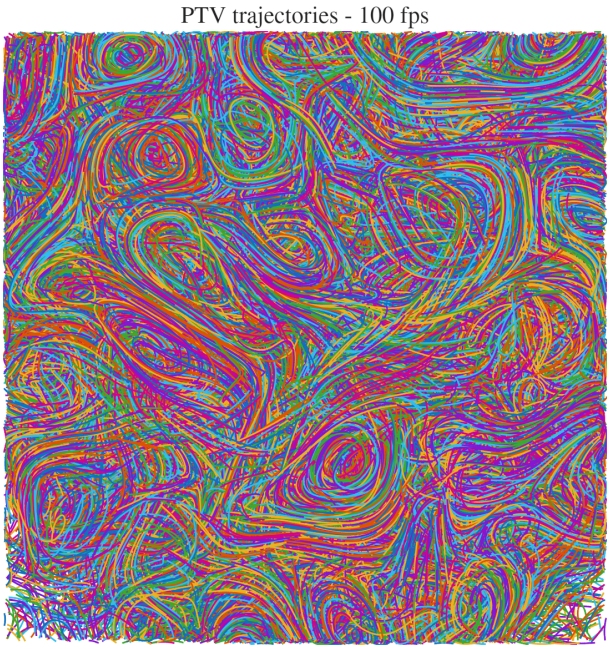


Figure 10: PTV Particle tracing

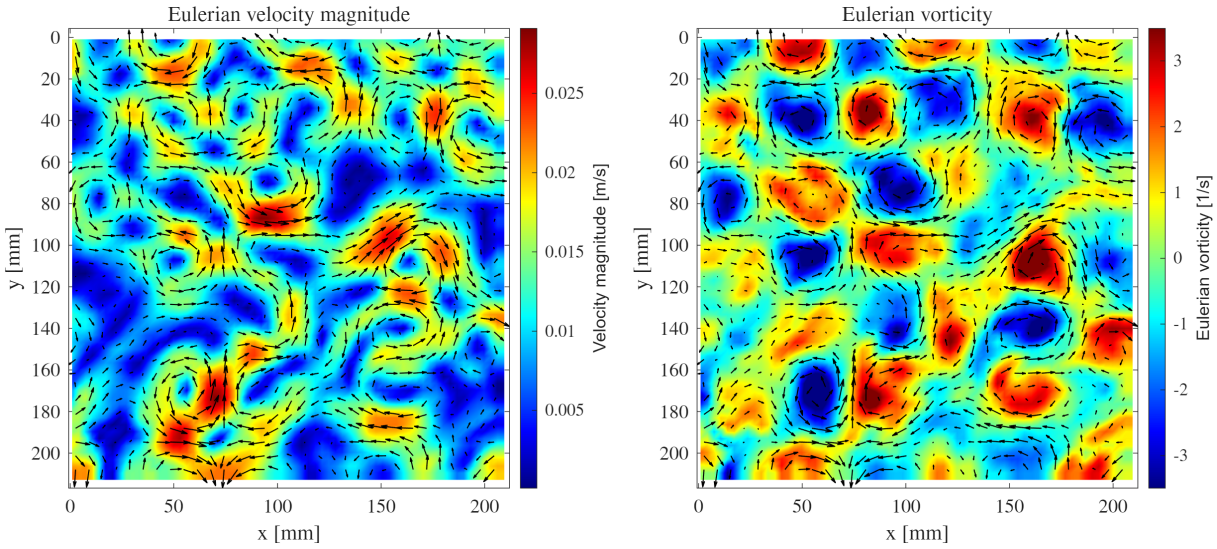


Figure 11: Eulerian conversion map of Vorticity and Magnitud

Third mode: 0.25A

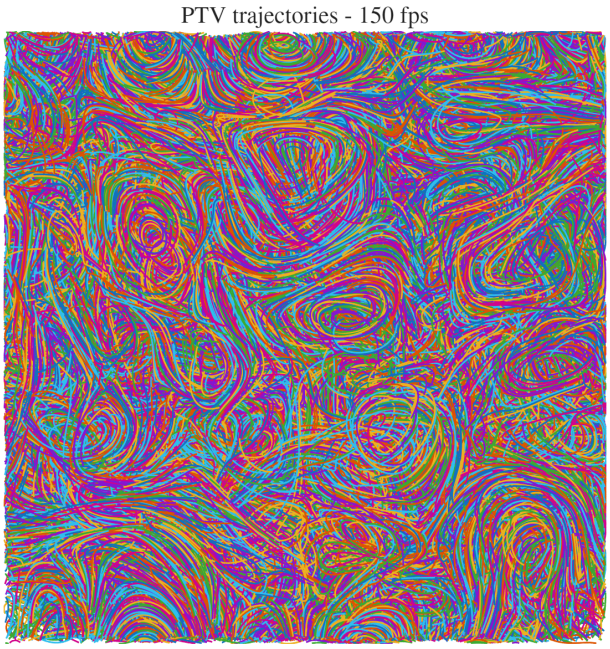


Figure 12: PTV Particle tracing

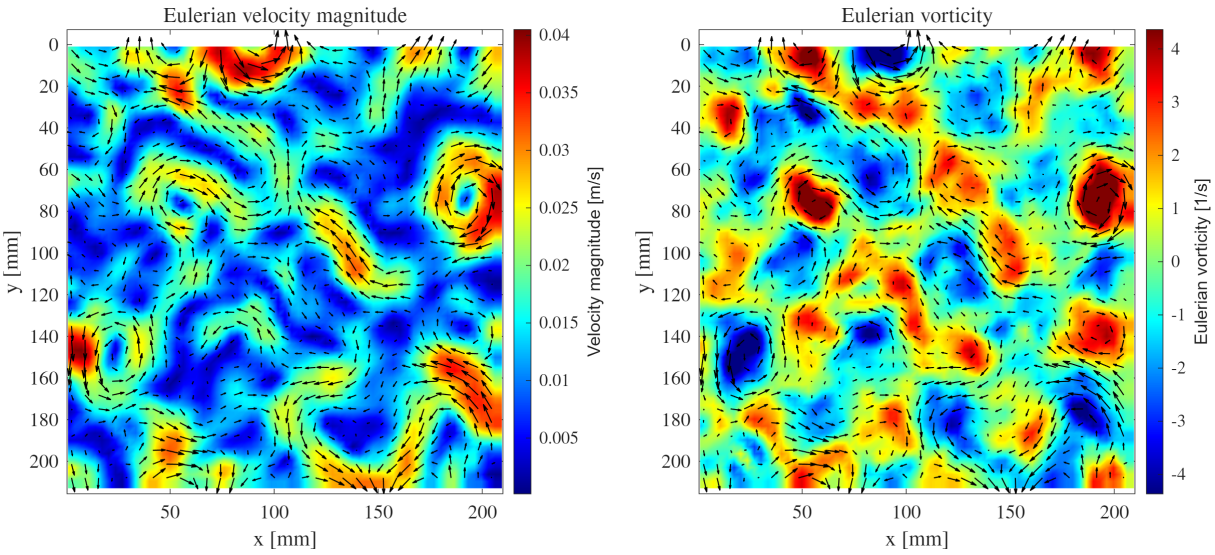


Figure 13: Eulerian conversion map of Vorticity and Magnitude

Fourth mode: 0.49A

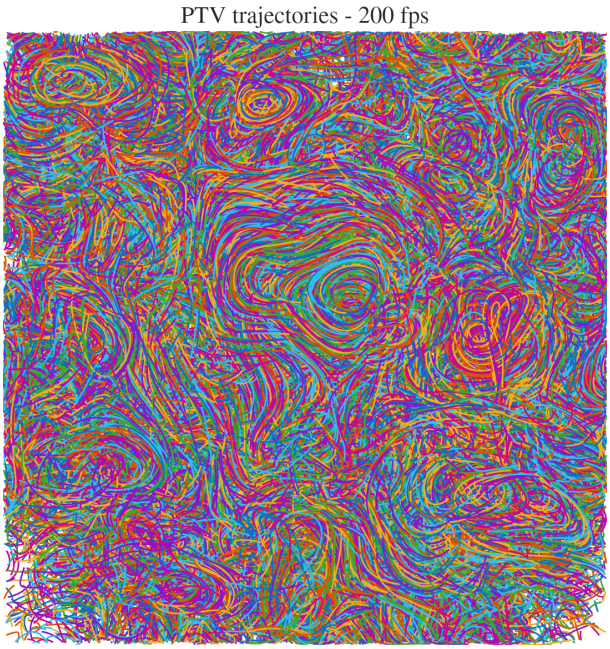


Figure 14: PTV Particle tracing

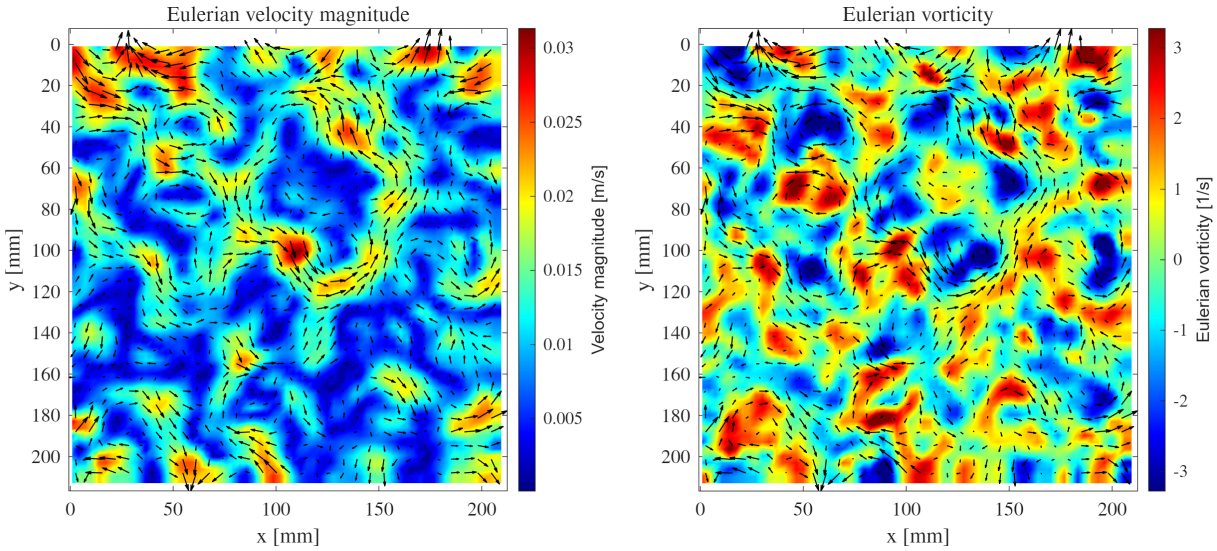


Figure 15: Eulerian conversion map of Vorticity and Magnitude

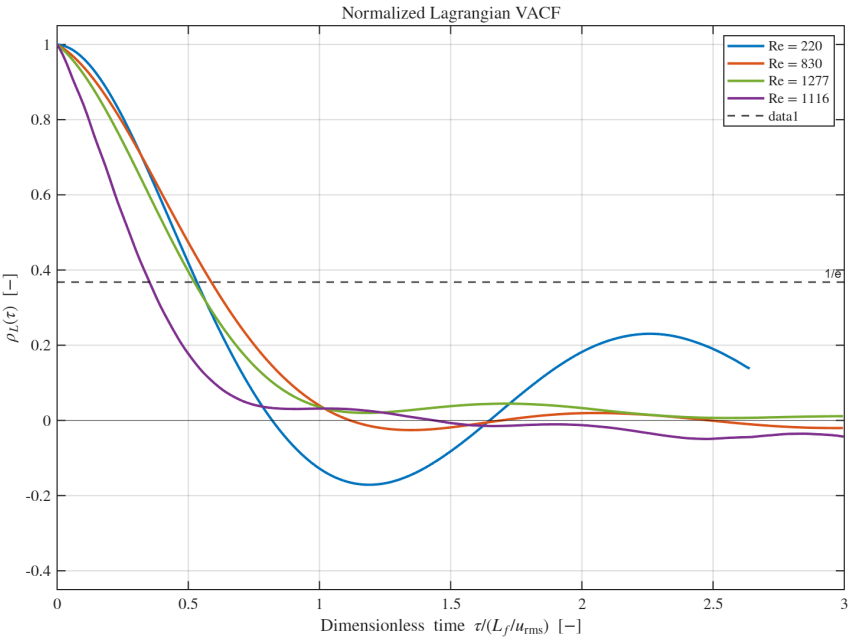


Figure 16: Normalized Lagrangian velocity autocorrelation for four Reynolds numbers.